



SW222 PROJECT REPORT: OVERHEAT PROTECTION CIRCUIT

Section: O42

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i. Abstarct.

In this project, we developed an Overheat Protection Circuit aimed at safeguarding electronic devices from the risks of overheating.

Our circuit utilizes a temperature sensor that continuously monitors the temperature of the device.

When the temperature exceeds a predefined limit, the circuit activates a fan or an alert system to prevent potential damage.

We used Verilog for the circuit design and simulated its functionality using Active-HDL.

The results showed that the circuit effectively responds to temperature changes, providing a practical solution for thermal management in electronic systems.

ii. introduction

Overheating is a common issue that can cause significant harm to electronic devices, leading to malfunctions and reduced performance. The aim of our project is to design an Overheat Protection Circuit that actively monitors and manages the temperature of devices. The circuit incorporates a temperature sensor to detect when the temperature rises above a safe threshold. Once this threshold is reached, the circuit activates a cooling fan or an alert system to prevent overheating.

By collectively designing this circuit, we aimed to enhance our understanding of electronic circuits and improve our skills in using simulation tools such as Active-HDL. This report will provide a detailed overview of the design process, the challenges we faced as a team, and the results of our simulations, demonstrating how effective thermal protection can extend the lifespan and reliability of electronic devices.

iii. Description and explanation.

The purpose of this project is to create a simple yet effective protection mechanism for electronic devices that may suffer from overheating. Overheating can cause performance degradation, hardware damage, or even system failure. To prevent this, our project introduces an Overheat Protection Circuit that continuously monitors temperature levels using a digital temperature input.

The system is implemented in Verilog HDL and simulated using Active-HDL. It receives a temperature value as an input and reacts based on predefined thresholds to control a cooling fan and an alarm. Unlike a simple on/off logic, our system applies multi-level control for better temperature management and safety:

- If the temperature is greater than or equal to 120°C:
 - The fan is activated.
 - The alarm is triggered.
- If the temperature is between 100°C and 119°C:
 - The fan is activated.
 - The alarm remains off.
- If the temperature is below 100°C:
 - The fan is turned off.
 - The alarm is turned off.

This approach improves safety and optimizes power usage by only activating outputs when necessary. The circuit is modular, cost-effective, and suitable for integration in systems such as processors, power electronics, or industrial equipment where thermal regulation is critical. It delivers a real-time response to overheating, helping prevent potential damage or failure.

iv. The source code .

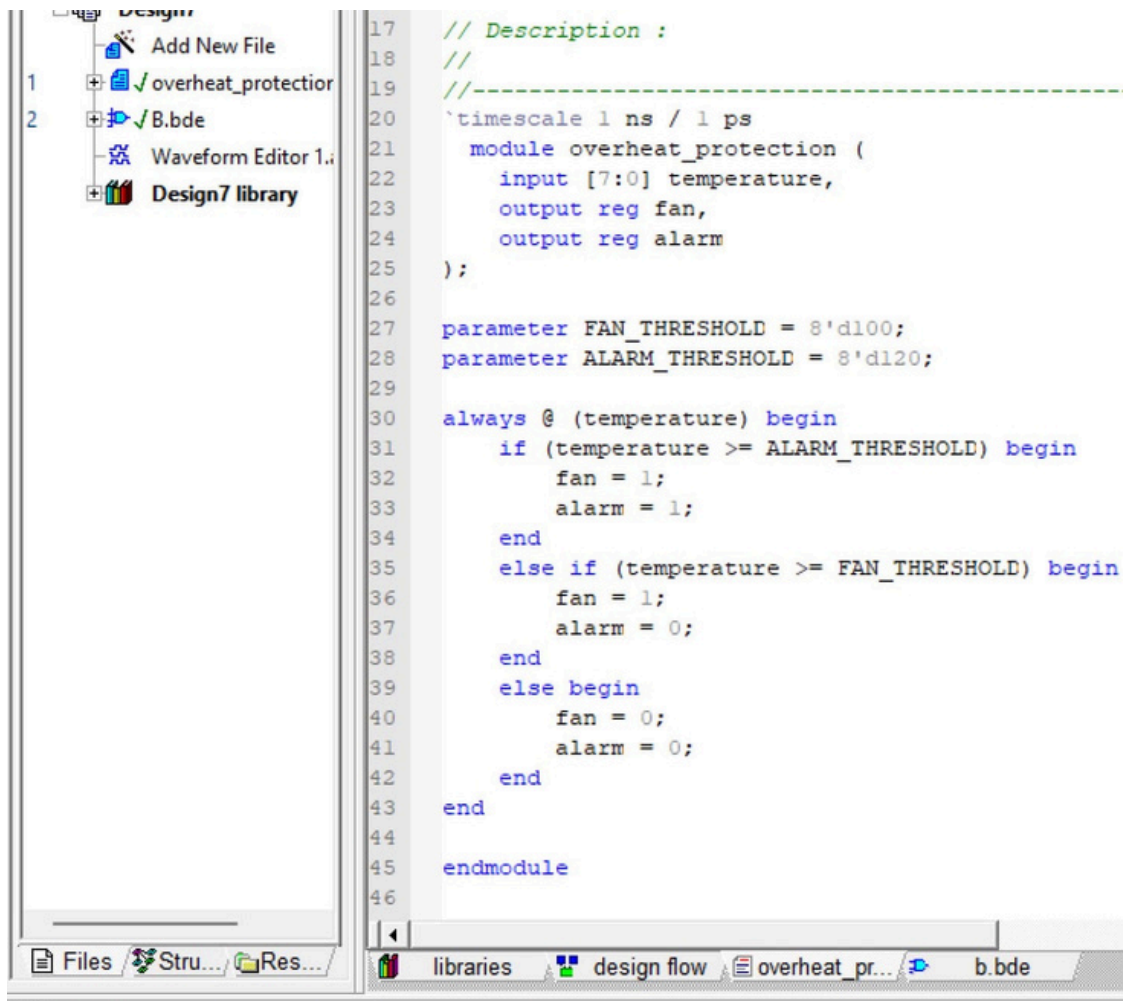
```
module overheat_protection (
    input [7:0] temperature, // 8-bit temperature input
    output reg fan,          // Fan control output (ON/OFF)
    output reg alarm         // Alarm control output (ON/OFF)
);

parameter fan_threshold = 8'd100; // Fan activation threshold
parameter alarm_threshold = 8'd120; // Alarm activation threshold

// Monitor the temperature and control outputs accordingly
always @ (temperature) begin
    if (temperature >= alarm_threshold) begin
        fan = 1; // Turn ON the fan
        alarm = 1; // Turn ON the alarm
    end
    else if (temperature >= fan_threshold) begin
        fan = 1; // Turn ON the fan
        alarm = 0; // Turn OFF the alarm
    end
    else begin
        fan = 0; // Turn OFF the fan
        alarm = 0; // Turn OFF the alarm
    end
end

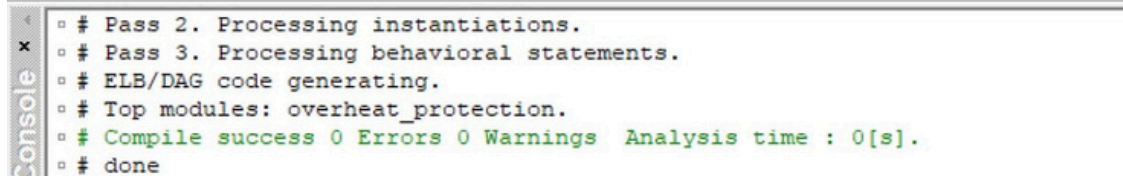
endmodule
```

v. Screenshot of the code and the diagram.



```
17 // Description :
18 //
19 //-----
20 `timescale 1 ns / 1 ps
21 module overheat_protection (
22     input [7:0] temperature,
23     output reg fan,
24     output reg alarm
25 );
26
27 parameter FAN_THRESHOLD = 8'd100;
28 parameter ALARM_THRESHOLD = 8'd120;
29
30 always @ (temperature) begin
31     if (temperature >= ALARM_THRESHOLD) begin
32         fan = 1;
33         alarm = 1;
34     end
35     else if (temperature >= FAN_THRESHOLD) begin
36         fan = 1;
37         alarm = 0;
38     end
39     else begin
40         fan = 0;
41         alarm = 0;
42     end
43 end
44
45 endmodule
46
```

The screenshot shows the code editor with the VHDL code for the 'overheat_protection' module. The left pane shows the project structure with 'overheat_protection' and 'B.bde' files. The bottom status bar indicates the current file is 'b.bde'.

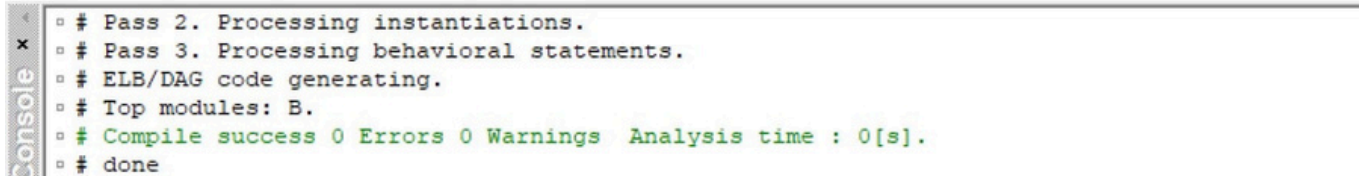
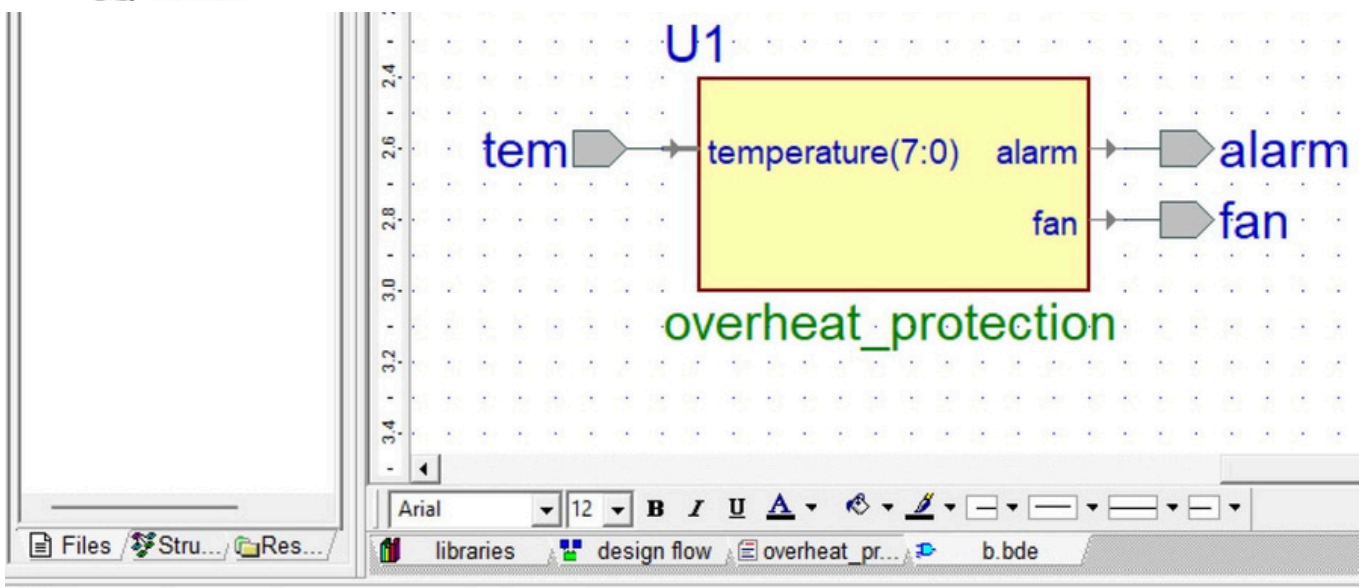


```

# Pass 2. Processing instantiations.
# Pass 3. Processing behavioral statements.
# ELB/DAG code generating.
# Top modules: overheat_protection.
# Compile success 0 Errors 0 Warnings Analysis time : 0[s].
# done

```

The console window displays the compilation progress, showing that the code was compiled successfully with no errors or warnings.



```

# Pass 2. Processing instantiations.
# Pass 3. Processing behavioral statements.
# ELB/DAG code generating.
# Top modules: B.
# Compile success 0 Errors 0 Warnings Analysis time : 0[s].
# done

```

The console window displays the compilation progress for the block diagram, showing that the code was compiled successfully with no errors or warnings.

vi. Screenshot of the simulation trace (waveform).

Name	Value	Stimulator ▾	20
temperature	82	<= 10000010	46
fan	1		
alarm	1		

Name	Value	Stimulator ▾	20	40	60	80
temperature	6E	<= 1101110	50			
fan	1					
alarm	0					

Name	Value	Stimulator ▾	20	40	60
temperature	46	<= 1000110	46		
fan	0				
alarm	0				

vii. References:

- Saber, M. J., & Ghorbani, M. (2019). "Design of an Overheat Protection System for Electronic Devices."
- Mano, M. Morris, and Michael D. Ciletti. *Digital Design with an Introduction to the Verilog HDL*. Pearson, 2012.
- "How to Design Over Temperature Protection Circuit." *Electronics Believer*, [How to Design Over Temperature Protection Circuit | electronics believer](#)
- "Designing a Verilog-based Temperature Sensor Interface for Alarm Triggers." *PeerDH*, <https://peerdh.com/blogs/programming-insights/designing-a-verilog-based-temperature-sensor-interface-for-alarm-triggers>